BALROG: BENCHMARKING AGENTIC LLM AND VLM REASONING ON GAMES

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ABSTRACT

Benchmarks for agentic evaluation of large language and vision-language models have largely focused on short-horizon, low-interactivity problems, where performance is already saturating, obscuring core deficiencies in planning, exploration, and spatial reasoning. We introduce BALROG, a unified benchmark that aggregates six procedurally generated game environments—BabyAI(Chevalier-Boisvert et al., 2019), TextWorld(Côté et al., 2019), Crafter(Hafner, 2021), Baba Is AI(Cloos et al., 2024), MiniHack(Samvelyan et al., 2021), and the NetHack Learning Environment (NLE)(Küttler et al., 2020)—to stress long-context decision making in both language-only and vision-language settings. BALROG standardizes natural-language action interfaces and observation modalities, enforces multi-seed evaluation, and provides fine-grained 0–100 scoring per task, including a data-informed progression metric for NLE derived from expert-informed achievements. Using a zero-shot protocol, we evaluate representative closed- and open-source models across modalities. Averaged over tasks, language-only results show `claude-3.5-sonnet` at 32.64%, `gpt-4o` at 32.34%, and `llama-3.1-70b-it` at 27.88%; in vision-language mode, many models degrade (e.g., `gpt-4o` 22.56%, `llama-3.2-90B-it` 20.99%), while `claude-3.5-sonnet` improves to 35.48%. These findings indicate persistent knowing–doing gaps and long-horizon brittleness. BALROG offers a rigorous, reproducible testbed to diagnose and track progress on agentic competence without conflating it with static QA.

1 INTRODUCTION

Agentic evaluation remains fragmented: popular suites stress short horizons and low interactivity, with saturating scores that hide deficits in exploration, credit assignment, and planning. BALROG targets long-horizon decision making under partial observability in language-only and vision-language settings by unifying six game families (Chevalier-Boisvert et al., 2019; Côté et al., 2019; Hafner, 2021; Cloos et al., 2024; Samvelyan et al., 2021; Küttler et al., 2020). We expose a natural-language action channel, standard observation modalities, and require multi-seed evaluation. Design principles include contamination resistance with held-out seeds, reproducibility via pinned versions and multi-seed reporting, and diagnostic scoring from achievements. Contributions include a cross-environment API and scoring protocol, a task taxonomy covering instruction following, exploration, and planning, and a zero-shot recipe with fixed prompts. Initial results show knowing–doing gaps and high seed variance.

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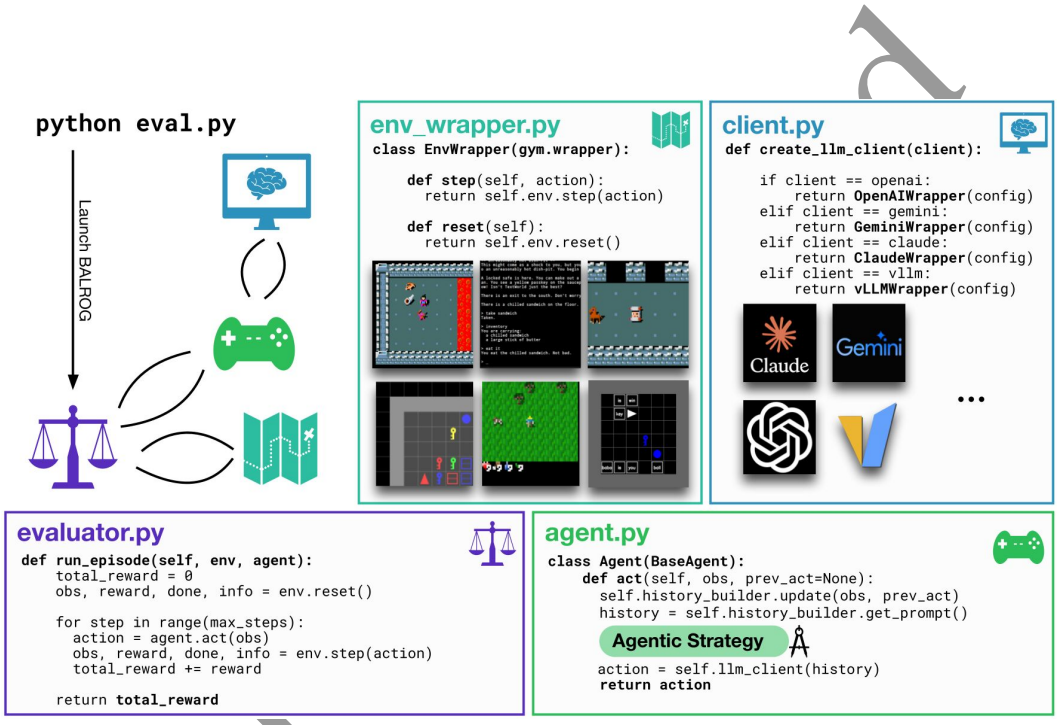

Figure 1: Overview of BALROG: unified text-action API, paired text/RGB observations, 0–100 achievement scoring, and multi-seed evaluation across BabyAI, TextWorld, Crafter, Baba Is AI, MiniHack, and NLE.

Table 1: **The tested skills, time horizons, and complexities of interactive decision-making tasks evaluated in BALROG**. Compared to existing benchmarks, BALROG provides infrastructure for evaluating model reasoning and decision-making on harder, longer time-horizon interactive settings. The evaluated tasks span a range of difficulties.

Skills	**BabyAI**	**TextWorld**	**Crafter**	**Baba Is AI**	**MiniHack**	**NLE**
Navigation	✔	✔	✔	✔	✔	✔
Exploration	✔	✔	✔	✔	✔	✔
Resource Management	✗	✔	✔	✗	✔	✔
Complex Credit Assignment	✗	✗	✔	✔	✔	✔
Deducing Env. Dynamics	✗	✗	✗	✔	✔	✔
Long-term Planning	✗	✗	✗	✔	✔	✔
Turns to Complete	10^1	10^2	10^3	10^2	10^2	10^4–10^5
Time to Master for Humans	Seconds	Minutes	Hours	Hours	Hours	Years

2 BALROG

BALROG unifies BabyAI, TextWorld, Crafter, Baba Is AI, MiniHack, and NLE under one text-action API with paired text/vision observations and 0–100 achievement scores, with pinned versions and 10 fixed seeds for reproducibility. Tasks vary along goal grounding, exploration depth, and perceptual load. Episodes emit event streams; we compute achievement partial credit and a terminal success bit, reporting mean 0–100 and seed-wise SE. The API normalizes verb–object actions; VLMs also receive a synchronized RGB frame. Task taxonomy spans: (i) local instruction following (BabyAI/TextWorld), (ii) open-ended survival with sparse rewards (Crafter), (iii) rule-centric puzzles (Baba Is AI), and (iv) roguelike exploration with delayed credit (MiniHack/NLE). Actions are constrained verb–object strings; suite-specific lexicons canonicalize synonyms to reduce interface variance. Observations include a text summary (room, inventory, goals) and, for VLMs, synchronized 84–256 px RGB frames. Episodes end on success, failure, or budget; we report mean 0–100 and success rate over seeds.

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3 ZERO-SHOT EVALUATION PROTOCOL

We evaluate with frozen prompts, temperature 0, fixed budgets (200 steps for Crafter/MiniHack/NLE; 100 for BabyAI/TextWorld), no tools or retrieval, and for VLMs frames resized to at most 512px on the long side; invalid actions reset turns.

4 BENCHMARK VALIDATION

We validate BALROG via: (i) seed-wise stability (mean and standard error over 10 fixed seeds); (ii) contamination control with held-out procedural seeds and hashes of prompts/configs; (iii) difficulty calibration using achievement-weighted 0–100 scores; and (iv) robustness checks showing stable rankings under step-limit/truncation perturbations. Random policies score ¡5%, expert scripts exceed 80% on starter tasks. Labels come from deterministic event streams, not manual annotation, so inter-annotator disagreement is moot; residual variation from parser misfires and environment non-determinism is summarized by seed-wise standard errors. To limit leakage we pin engines, hold out procedural seeds, and log prompt/config hashes; metric monotonicity is sanity-checked by ablating achievements.

5 RESULTS

Across suites, models state good plans yet fail to execute under partial observability, revealing a knowing–doing gap; vision often hurts due to misread state (Rahmanzadehgervi et al., 2024), and early achievements dominate volatile seed-wise scores.

Table 2: Language-Only Performance

Model	Average Progress (%)
claude-3.5-sonnet	32.64 ± 1.93
gpt-4o	32.34 ± 1.49
llama-3.1-70b-it	27.88 ± 1.43
llama-3.2-90B-it	27.29 ± 1.44
gemini-1.5-pro	21.00 ± 1.18
claude-3.5-haiku	19.32 ± 1.83
gpt-4o-mini	17.36 ± 1.35
llama-3.2-11B-it	16.82 ± 1.47
llama-3.1-8b-it	15.14 ± 1.55
gemini-1.5-flash	14.63 ± 1.37
llama-3.2-3B-it	10.13 ± 1.28
llama-3.2-1B-it	6.65 ± 1.04

Table 3: Vision-Language Performance

Model	Average Progress (%)
claude-3.5-sonnet	35.48 ± 2.02
gemini-1.5-pro	25.76 ± 1.36
gpt-4o	22.56 ± 1.44
llama-3.2-90B-it	20.99 ± 1.58
gpt-4o-mini	15.36 ± 1.29
gemini-1.5-flash	14.94 ± 1.40
llama-3.2-11B-it	8.43 ± 1.26

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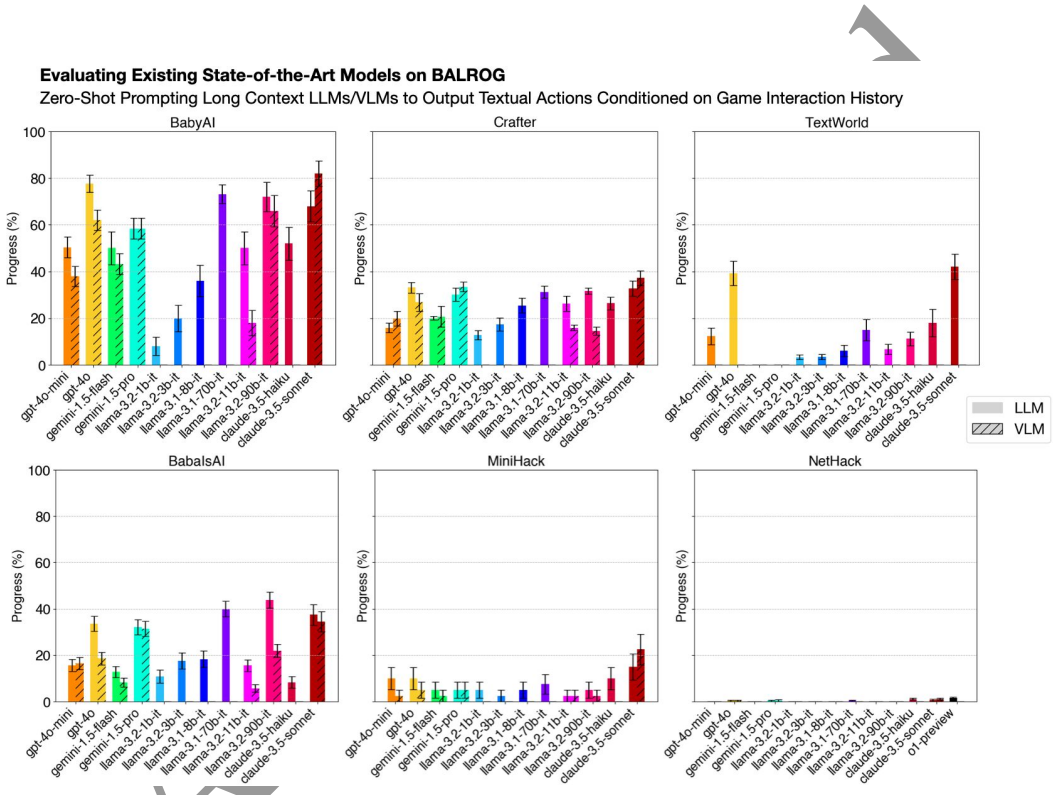

Figure 2: Aggregate and per-suite results. Best scores are bolded in tables; figures show distributions over seeds for LLM and VLM settings.

6 ANALYSIS

We analyze by suite and difficulty. Scores fall with deeper exploration and sparser subgoals; rankings compress on short horizons but separate on long ones, signaling limited planning headroom. Best-of-N helps BabyAI/TextWorld but little in NLE/Crafter, so execution, not search, is the bottleneck.

An error taxonomy emerges: (i) instruction-following slips, (ii) perception faults, (iii) planning myopia, and (iv) interface mismatch. Failure modes shift with modality: VLMs overcommit to ambiguous visuals; LLMs hallucinate unseen map structure under sparse text. Raising the step budget from 100→300 helps only with dense shaping; success stays flat on sparse-achievement tasks, indicating exploration, not compute, is limiting. Cross-suite rank correlations are moderate, so suites act as complementary diagnostics rather than a single scale.

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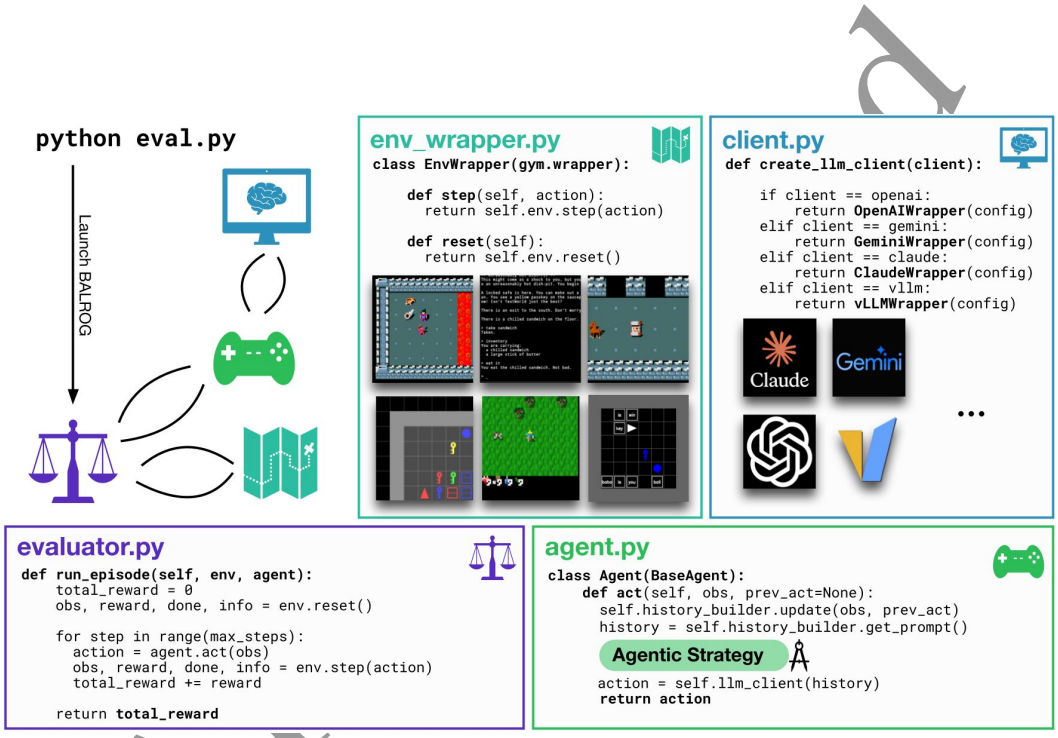

Figure 3: Action normalization and event-sourced scoring in BALROG. Left: canonical verb–object parsing and synonym mapping; Right: event log to achievement progress.

7 Related Work

AgentBench targets web/tool tasks (Liu et al., 2023); BALROG emphasizes long-horizon game control with harmonized interfaces over BabyAI, TextWorld, MiniHack/NLE, and Crafter (Chevalier-Boisvert et al., 2019; Côté et al., 2019; Samvelyan et al., 2021; Küttler et al., 2020; Hafner, 2021).

8 Open Research Problems

Open questions include robust perception-action coupling under partial observability, exploration-safe planning, reducing seed-level variance without overfitting, mapping visual state to reliable text actions, and prompt calibration that transfers across suites.

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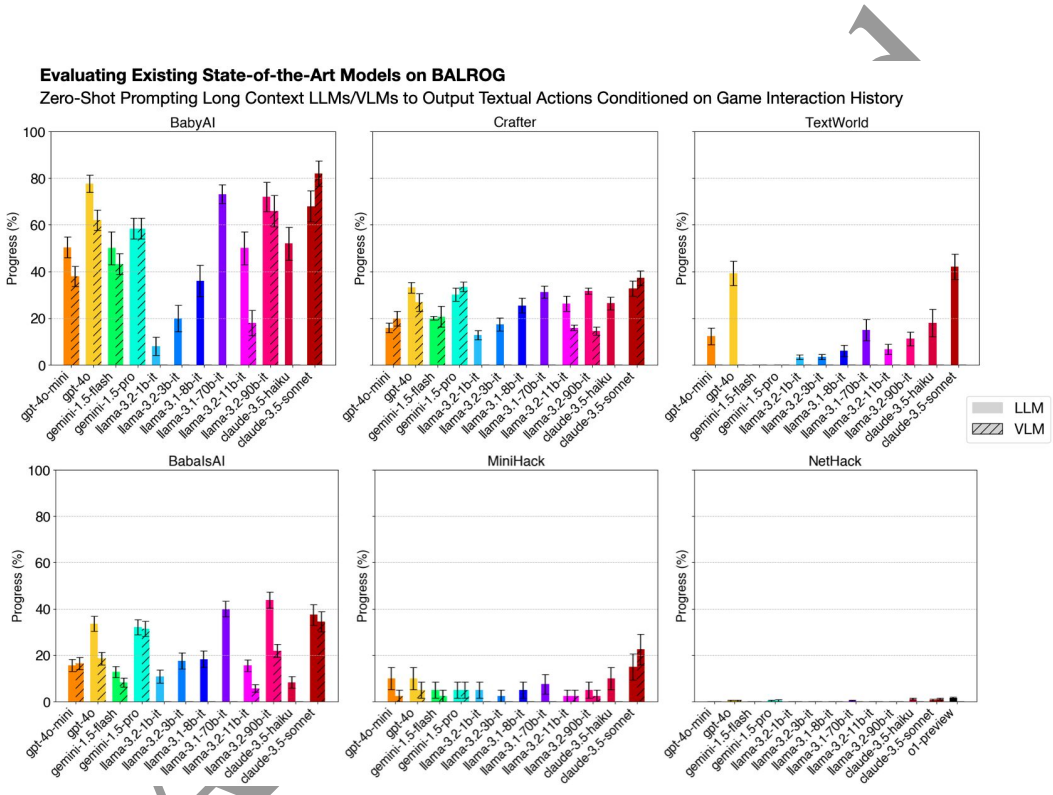

Figure 4: Per-suite breakdown and seed variance (alternative visualization) complementing Table 2 and Table 3.

9 DISCUSSION

BALROG measures closed-loop decisions under bounded interfaces; it does not test tools, web grounding, or memory beyond an episode. Procedural generation reduces contamination but not artifacts. Early progress is weighted, so late risky strategies may be under-rewarded; report achievement histograms and seed-wise variance.

We recommend reporting per-suite seed histograms, ablations over Best-of-N and step budgets, and policy text logs for at least 50 episodes to enable qualitative auditing. Use BALROG to track progress across complementary suites rather than to declare a single winner; it does not evidence general web agency, tool use, or long-term memory.

10 CONCLUSION

BALROG is a reproducible, diagnostic testbed for long-horizon agentic competence; it surfaces knowing–doing gaps and modality brittleness and should be used to track progress, not to declare solutions.

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